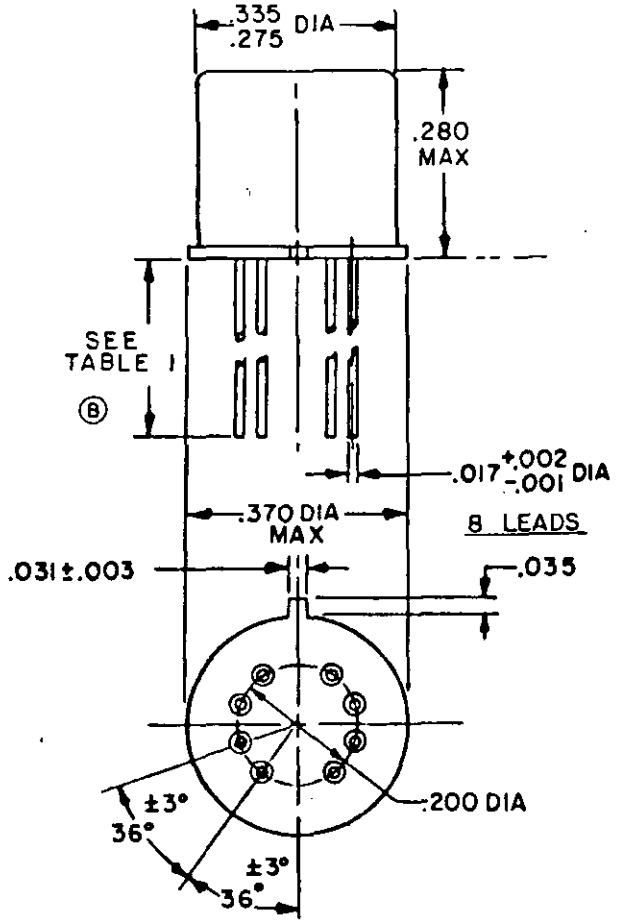
MILITARY SPECIFICATION SHEET

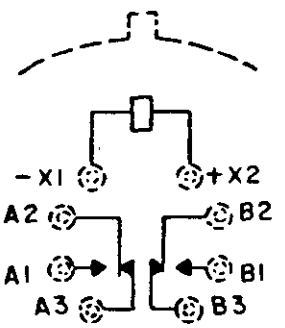
RELAYS, ELECTROMAGNETIC, HIGH SHOCK AND HIGH VIBRATION, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 1.0 AMPERE

Ⓑ This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-39016.



Inches	mm
.001	0.03
.002	0.05
.003	0.08
.017	0.43
.031	0.79
.035	0.89
.200	5.08
.275	6.99
.280	7.11
.335	8.51
.370	9.40



CIRCUIT DIAGRAM
TERMINAL VIEW

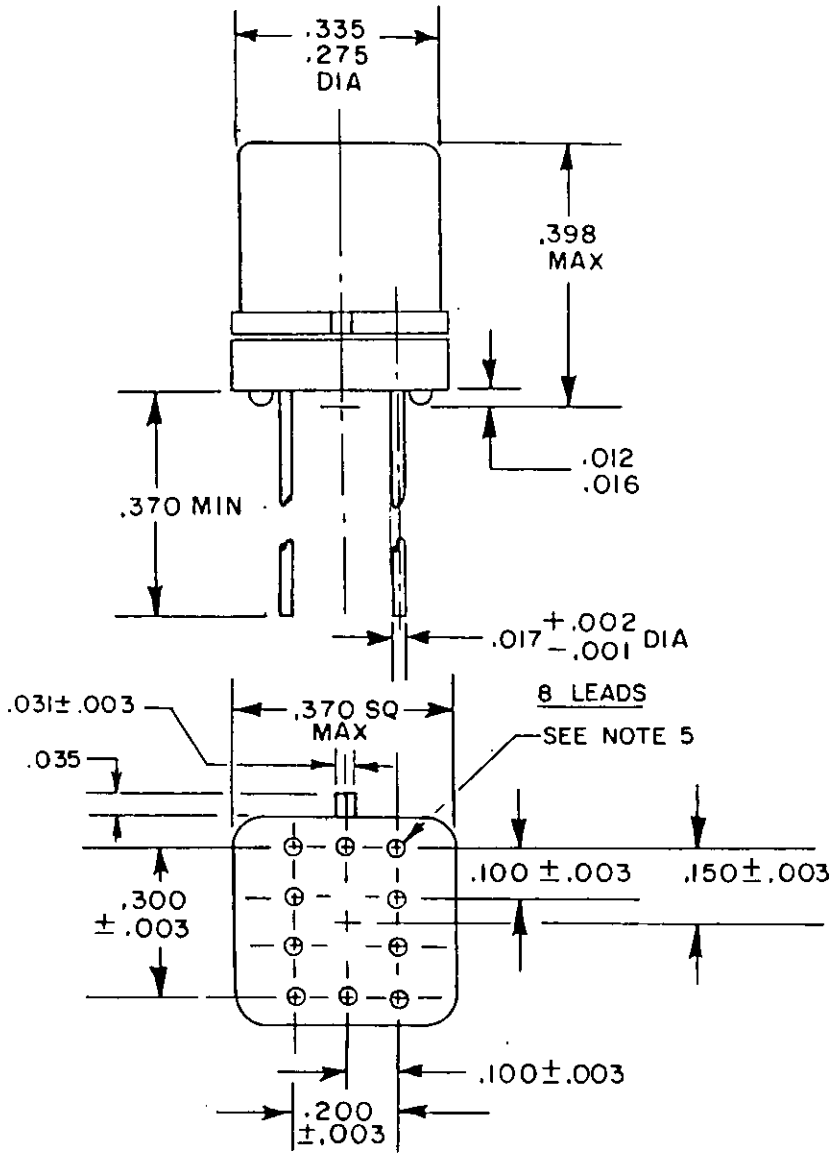
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above for reference only. Numbers do not appear on relay.
5. Coil symbol optional in accordance with MIL-STD-1285.
6. Circuit diagram shown on part is the terminal view.
7. Relays shall have a (+) sign placed on the circuit diagram as shown.

FIGURE 1. Dimensions and configuration.

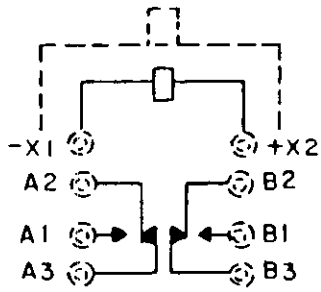
Ⓑ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:52Z
Check the source to verify that this is the current version before use.



Inches	mm
.001	0.03
.002	0.05
.003	0.08
.012	0.30
.016	0.41
.017	0.43
.031	0.79
.035	0.89
.100	2.54
.150	3.81
.200	5.08
.275	6.99
.300	7.62
.335	8.51
.370	9.40
.398	10.11



CIRCUIT DIAGRAM
TERMINAL VIEW

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Spreader pads shall be certified to MIL-M-38527, M38527/5-003 or -013.
5. Dimensions and tolerances shown for the bottom view of the spreader pad are for the center to center locations of the holes in the spreader pad.
6. Shape optional within envelope dimension.
7. Coil symbol optional in accordance with MIL-STD-1285.
8. Terminal numbers shown above for reference only. Numbers do not appear on relay.
9. Circuit diagram shown on part is the terminal view.
10. Relays shall have a (+) sign placed on the circuit diagram as shown.

FIGURE 2. Dimensions and configuration relays supplied with spreader pad.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:52Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

CONTACT DATA

Load ratings:

High level (relay case grounded):

Resistive:

1.0 ampere at 28 V dc.
500 milliamperes at 115 V ac 400 Hz case not grounded.
250 milliamperes at 115 V ac 60 Hz case not grounded.
100 milliamperes at 115 V ac 60 and 400 Hz case grounded.

Inductive load: 0.2 ampere at 28 V dc with 0.32 henry inductance.

Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

(B) Initial: 0.10 ohm maximum (0.125 ohm maximum with spreader pad attached).

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.20 ohm maximum (0.225 ohm maximum with spreader pad attached).

Low level:

During life: 100 ohms maximum.

After life: 0.15 ohm maximum (0.175 ohm maximum with spreader pad attached).

Intermediate current:

During intermediate current: 3 ohms maximum.

After intermediate current: 0.20 ohm maximum (0.225 ohm maximum with spreader pad attached).

Contact bounce: 1.5 milliseconds maximum (applicable to failure rate level "L").

Contact stabilization time: 2.0 milliseconds maximum (applicable to failure rate levels "M", "P", and "R").

Overload (high level only): Two times rated current.

COIL DATA (see table I)

Operate time: 6.0 milliseconds maximum over temperature range with rated coil voltage.

Release time: 3.5 milliseconds maximum over temperature range from rated coil voltage.

Coil life: Applicable. (For retention of qualification testing, coil life shall be performed following vibration testing in group II of the group C inspection table of MIL-R-39016. Test time shall be 250 hours.)

Source: https://assist.dla.mil -- Downloaded: 202610-21T17:52Z
Check the source to verify that this is the current version before use.

ELECTRICAL DATA

Insulation resistance: 10,000 megohms minimum at 500 V dc, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage:

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts both in the energized and deenergized positions	500	125 All terminals to case
Between case, frame, or enclosure and coil(s)	500	
Between all contacts and coil(s)	500	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	500	

ENVIRONMENTAL DATA

Temperature range: -65°C to +125°C.

Vibration: Applicable, except vibration level shall be 100 g's and frequency range shall be 140 to 2,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

(B) Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts (applicable to qualification and group C testing only).

Shock (specified pulse): MIL-STD-202, method 213, test condition B, except peak value shall be 150 g's for 11 milliseconds. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): In accordance with method 1041, MIL-STD-750.

PHYSICAL DATA

Terminals:

Terminal strength: Test condition A, 1 pound pull minimum.

Bend test: Test condition C, 1/2 pound load.

(B) Terminal twist test: As specified in MIL-R-39016.

(B) Solderability: Applicable.

Dimensions and configuration: See figures 1 and 2.

Terminations: See figures 1 and 2 and table I.

Weight: 3.1 grams (0.109 ounce) maximum (3.66 grams (0.129 ounce) with spreader pad attached).

Source: https://assist.dla.mil -- Downloaded: 202610-21T17:52Z
Check the source to verify that this is the current version before use.

Seal: Hermetic.

Internal moisture: Applicable.

Ⓑ Minimum marking: Military part number, "J" with the date code (example J8530), circuit diagram, manufacturer's name or source code.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per relay.

Ⓑ Low level: 100,000 cycles plus 900,000 cycles mechanical life.

Part number: M39016/49- (dash number from table I and suffix letter designating failure rate level).

Ⓑ TABLE I. Dash numbers and characteristics. 1/

Dash numbers 2/ 3/			Coil voltage (V dc) 4/		At +25°C				Over temperature range		
Lead length .500 min	Lead length .187 +.040 -.010	Spreader pads (figure 2) 3/	Rated	Max	Coil resistance ohms ±10%	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified drop-out value (voltage) (V dc)	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified drop-out value (voltage) (V dc)
001	007	013	5.0	5.8	50	3.5	1.4	0.22	4.6	2.5	0.14
002	008	014	6.0	8.0	70	4.2	2.0	0.27	5.5	3.2	0.18
003	009	015	9.0	12.0	155	6.2	3.0	0.54	8.2	4.9	0.35
004	010	016	12.0	16.0	235	8.3	4.0	0.63	11.0	6.5	0.41
005	011	017	18.0	24.0	610	12.5	6.0	0.77	16.5	10.0	0.59
006	012	018	26.5	32.0	1,130	18.0	8.0	1.23	22.0	13.0	0.89

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 100,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example: 001L - - - - - - 012R.

3/ Relays supplied with spreader pads (-013 through -018) shall have the pad rigidly attached.

4/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

Source: https://assist.dla.mil -- Downloaded: 202640-21T17:52Z
Check the source to verify that this is the current version before use.

QUALIFICATION INSPECTION

Qualification inspection and sample size: See table II.

(B) TABLE II. Qualification inspection and sample size. 1/

Single submission	Group submission	
18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable	M39016/49-006	18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable
	M39016/49-001 M39016/49-002 M39016/49-003 M39016/49-004 M39016/49-005	2 units each part number Qualification inspection, group II

1/ For retention or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-39016/9 or MIL-R-39016/50 may be used in addition to MIL-R-39016/49 data. Prior to performance of retention of qualification testing; the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing will be increased as required in group V, table II, MIL-R-39016, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to performance of qualification inspection testing; the relay manufacturer shall preselect the sampling plan.

Initial qualification of relays supplied with spreader pads (-013 through -018) shall be tested as specified below:

(B) Perform the following tests as specified in the qualification inspection table of MIL-R-39016, in the order shown before installation of pads; screening, visual and mechanical examination (internal), thermal shock, resistance to solvents, vibration (sinusoisdal), vibration (random), shock (specified pulse), acceleration, terminal strength, magnetic interference (when specified), capacitance (when specified), coil life (applicable to continuous duty relays only), resistance to soldering heat, salt spray (corrosion), overload (applicable to high level relays only), life, terminal strength, and intermediate current.

(B) After installation of pads perform the following tests as specified in the qualification inspection table of MIL-R-39016, in the order shown; insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, and visual and mechanical inspection (external).

Qualification inspection (reduced testing for previously qualified relays) for relays supplied with spreader pads (-013 through -018), 2 units of the 26.5 volt rated coil voltage (-018) shall be tested as specified below:

Before installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016 in the order shown:

For failure rate level L only: Screening.

(B) For failure rate levels M, P, and R: Vibration (sinusoidal) test duration shall be 10 minutes vibration (random); particle impact noise detection (PIND, when specified), screening.

Source: https://assist.dla.mil -- Downloaded: 202610-21T17:52Z
Check the source to verify that this is the current version before use.

After installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016 in the order shown:

(B) Insulation resistance, dielectric withstanding voltage, static contact resistance; specified pickup, hold, and dropout values (values), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, and visual and mechanical inspection (external).

Group A testing for relays supplied with spreader pads (-013 through -018), shall be tested as specified below:

(B) Before installation of pad perform subgroup 2 of group A tests.

(B) After installation of pad perform subgroups 3 and 4 of group A tests.

Qualification inspection (reduced testing) and sample size: See table III.

If the relays produced for MIL-R-39016/49 are similar in construction and design except for the diode and coil assembly (as applicable) to the relays produced for MIL-R-39016/9 or MIL-R-39016/50, then reduced testing for qualification of MIL-R-39016/49 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-39016/9 or MIL-R-39016/50. For reduced testing, see table III.

(B) TABLE III. Qualification inspection (reduced testing).

Examination or test
2 units tested as specified below: 26.5 V dc coil voltage Group II of qualification inspection table Shock Vibration (as specified for qualification) Life All other coil voltages Group II of qualification inspection table

CONCLUDING MATERIAL

Custodians:
(B) Army - ER
(B) Navy - EC
(B) Air Force - 85

User activity:
(B) Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5945-0757-41)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:52Z
Check the source to verify that this is the current version before use.